

**ASSESSMENT OF RAILROAD BALLAST DEGRADATION UNDER
MODIFIED DROP-WEIGHT IMPACT LOADING CONDITIONS
AND IMAGE-BASED MORPHOLOGICAL ANALYSIS**

A Thesis

Submitted to the School of Technology

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Gati Shakti Vishwavidyalaya

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Candidate's Declaration

We hereby certify that:

- (i) The work contained in this thesis is original and has been carried out by us jointly under the guidance of our supervisor, Dr. Dinesh Gundavaram, Gati Shakti Vishwavidyalaya, Vadodara.
- (ii) The work has not been submitted in whole or in part to any other institute or university for the award of any degree, diploma, or certificate.
- (iii) We have followed the academic guidelines and ethical code of conduct provided by Gati Shakti Vishwavidyalaya in preparing this thesis.
- (iv) Wherever we have used materials, data, experimental results, theoretical analyses, figures, tables, or text from other sources, we have given due credit to them by citing them in the text of the thesis and by giving full bibliographic details in the reference list. Where required, permission from the copyright owners has been obtained.
- (v) The experimental programme reported in this thesis was conducted by both candidates jointly at Gati Shakti Vishwavidyalaya.

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Certificate of Thesis Supervisor

This is to certify that the thesis entitled “*Assessment of Railroad Ballast Degradation under Modified Drop-Weight Impact Loading Conditions and Image-Based Morphological Analysis*”, submitted by Durgesh Kumar Legha (Roll No. 2270007) and Sandeep Patel (Roll No. 2280043) to Gati Shakti Vishwavidyalaya, is a record of bonafide research work carried out jointly by them under my supervision. The work embodied in this thesis has not been submitted, in whole or in part, for the award of any degree, diploma, or certificate at any other institute or university. In my opinion, this thesis is of the standard required for the degree of Bachelor of Technology in Civil Engineering (Specialisation: Railway Engineering), and I consider it worthy of consideration for the said degree.

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Certificate of Approval

Certified that the thesis entitled “*Assessment of Railroad Ballast Degradation under Modified Drop-Weight Impact Loading Conditions and Image-Based Morphological Analysis*” submitted by Durgesh Kumar Legha (Roll No. 2270007) and Sandeep Patel (Roll No. 2280043) to Gati Shakti Vishwavidyalaya for the award of the Degree of Bachelor of Technology in Civil Engineering (Specialisation: Railway Engineering) has been accepted by the B.Tech. project committee members. The project presentation by the candidates was held on 23rd April 2026.

Dr. Dinesh Gundavaram
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B.Tech. Programme Coordinator

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Durgesh & Sandeep

Abstract

This thesis addresses the critical issue of railway ballast degradation, which significantly impacts the safety, stability, and operational efficiency of railway tracks. Tailored specifically to the unique conditions of Indian Railways, the research aims to develop a deeper understanding of ballast behaviour under impact loading and to quantify the magnitude of ballast degradation using three-dimensional image processing.

At the core of the study is the modification of the European-standard drop-weight impact-test apparatus to better align with Indian Railways ballast specifications and operational conditions. The experimental set-up is calibrated so that it accurately replicates the on-field stress conditions experienced by ballast in the Indian context, providing a more relevant and reliable framework for analysis. The modified apparatus serves as a critical tool for evaluating ballast degradation under controlled yet realistic conditions.

A significant innovation in this study is the use of advanced three-dimensional image-processing techniques to evaluate morphological changes in ballast particles before and after impact testing. High-resolution structured-light scans are captured to analyse changes in particle shape, size, and texture, providing a detailed and quantitative understanding of ballast degradation. This paired before–after approach allows precise measurement of deterioration, enabling the identification of critical thresholds and patterns of degradation that are not easily detectable through traditional methods.

The outcomes of this study advance railroad-ballast degradation assessment through image-based morphology, delivering a scalable and accurate complement to conventional physical testing, with substantial implications for railway safety and maintenance.

Keywords: railway ballast; drop-weight impact loading; Indian Railways; fouling index; structured-light scanning; three-dimensional particle morphology; under-ballast mat.

Outcomes of This Project

Journal Papers

1. Legha, D., & Gundavaram, D. *Composite Railway Sleepers: A Review of Mechanical Performance, Durability and Adoption Challenges*. Railway Engineering Science (2026). (Under Review). (IF = 5.4)
2. Legha, D. K., Gundavaram, D., & Patel, S. *A Large-Scale Drop-Weight Impact-Testing Apparatus for Assessment of Indian Railway Ballast: Influence of Stress Level and Subgrade-Interlayer Configuration on Settlement and Particle Degradation*. Construction and Building Materials (2026). (Under Review). (IF = 7.2)

Patents

1. Gundavaram, D., & Legha, D. (2026). *Assessment of Railroad Ballast Degradation under Modified Drop-Weight Impact Loading Conditions*. (Indian Utility Patent) (Draft Prepared).

Conference Papers

1. Legha, D. K., Patel, S., & Gundavaram, D. (2026). *Optimization of Bitumen Content in Bitumen-Stabilized Ballast Using Flowability Characteristics*. Recent Research on Environmental Earth Sciences, MedGU 2025. Springer. (**Accepted, In Press**)
2. Legha, D., Gundavaram, D., & Mishra, G. (2026). *Global Practices of Railway Drainage Systems with Emphasis on India and Emerging Innovation*. Geotechnical Practices for Innovation and Sustainability, Proceedings of the Indian Geotechnical Conference (IGC 2025), Lecture Notes in Civil Engineering, Springer. (**Accepted, In Press**)



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List of Abbreviations

Abbreviation	Full form / Description
AAV	Aggregate Abrasion Value
AIV	Aggregate Impact Value
ANOVA	Analysis of Variance
BIS	Bureau of Indian Standards
BTP	B.Tech. Project
CAD	Computer-Aided Design
$D/d_{\max}$	Ratio of mould diameter to maximum particle size (boundary criterion)
DEM	Discrete Element Method
EN	European Norm / European Standard
FB	Flexible Base (100 mm compacted sand layer, present study)
FI	Fouling Index (after Selig & Waters, 1994)
FI _{IN}	Indian Railway Fouling Index
GSV	Gati Shakti Vishwavidyalaya
HH	High Height (450 mm drop) — legacy laboratory label
IPO	Indian Patent Office
IQR	Inter-Quartile Range
IRJ	Insulated Rail Joint
IS	Indian Standard
LAA	Los Angeles Abrasion
LH	Low Height (300 mm drop) — legacy laboratory label
Mode A	Diffuse corner abrasion (present-study classification)
Mode B	Catastrophic single-particle bulk fracture (present-study classification)
PUBM	Polyurethane Under-Ballast Mat (Getzner, present study)
$P_{0.075}$	Percentage finer than the 0.075 mm sieve
P_{20}	Percentage finer than the 20 mm sieve

Abbreviation	Full form / Description
PSD	Particle Size Distribution
RB	Rigid Base (concrete mould floor, present study)
RDSO	Research Designs and Standards Organisation
RUBM	Rubber Under-Ballast Mat (12 mm, present study)
SciPy	Scientific Python library
SEM	Scanning Electron Microscopy
SL	Structured-Light (scanning)
STL	Stereolithography (triangular surface-mesh file format)
UBM	Under-Ballast Mat
UIC	Union Internationale des Chemins de fer
UR	Unreinforced (no pad, present study)
USP	Under-Sleeper Pad

List of Symbols

Symbol	Meaning (units)
A	Surface area of a particle (mm^2)
A_{ch}	Convex-hull surface area (mm^2)
a, b, c	Semi-axes of best-fit ellipsoid (mm), $a \geq b \geq c$
C	Convexity = V/V_{ch} (dimensionless)
D	Mould internal diameter (mm)
D_{eq}	Equivalent spherical diameter = $(6V/\pi)^{1/3}$ (mm)
d_{max}	Maximum particle size in a gradation (mm)
E	Elongation = I/L (dimensionless)
F	Flatness = S/I (dimensionless)
g	Gravitational acceleration, 9.81 m s^{-2}
h	Drop height of hammer (mm or m)
I	Intermediate bounding-box dimension (mm)
K_{e}	Kinetic energy of hammer at impact = mgh (J)
L	Long bounding-box dimension (mm)
L/S	Aspect ratio (dimensionless)
$(L + I)/(2S)$	Flatness index (dimensionless)
m	Hammer mass (kg)
N	Number of blows (dimensionless)
n	Sample size (dimensionless)
p	Statistical p -value (dimensionless)
R	Roundness descriptor (dimensionless, scanner-specific)
S	Short bounding-box dimension (mm)
t	Student's t -statistic (dimensionless)
V	Volume of a particle (mm^3)
V_{ch}	Convex-hull volume (mm^3)

Symbol	Meaning (units)
σ	Nominal contact stress at specimen surface (kPa)
ΔX	Change in descriptor $X = X_{\text{after}} - X_{\text{before}}$
$\% \Delta X$	Percentage change in $X = \Delta X / X_{\text{before}} \times 100$
Ψ	Sphericity = $\pi^{1/3} (6V)^{2/3} / A$ (Wadell, 1932)

CHAPTER 1

Introduction

1.1 Background: the central role of ballast in railway track

Railway ballast is a fundamental component of conventional track structures, consisting of coarse, angular aggregates—typically crushed stone—placed beneath and around sleepers to ensure track stability and performance. Its primary functions include distributing train loads from the sleepers to the underlying subgrade, providing lateral and longitudinal resistance to track movement, and facilitating effective drainage to prevent water accumulation and loss of strength in the track foundation (Navikas et al., 2018). The mechanical behaviour of ballast is governed by factors such as particle size, shape, gradation, and strength, which influence its load-bearing capacity and resistance to degradation under repeated train loading (Koozmishi & Palassi, 2017). Over time, ballast experiences fouling and particle breakage, which can reduce its drainage capability and stiffness, leading to track settlement and increased maintenance requirements (Bassey et al., 2020). Recent studies emphasise the importance of understanding the geomechanical properties and dynamic response of ballast to improve track longevity and safety, especially under high-speed and heavy-haul conditions (Alabbasi and Hussein, 2021). Consequently, ongoing research explores alternative materials, stabilisation techniques, and advanced monitoring methods to enhance ballast performance and sustainability in modern railway systems.

The mechanical behaviour of the ballast layer is governed, ultimately, by the morphological state of its constituent grains. Freshly quarried angular particles possess interlocking micro-asperities that generate very high inter-particle friction; a well-compacted layer of such particles can carry train loads with lateral deformations of less than a millimetre per million-gross-tonne of traffic. The same layer, after a decade of service, contains particles that have been rounded by abrasion and reduced in size by fracture. The rounded assembly has lower friction, the fractured fines infill the voids and reduce drainage, and the layer settles more quickly under each successive load cycle. Track geometry degrades, the track engineer dispatches a tamping machine, and a maintenance cycle begins. Esveld (2001), Lichtberger (2005), and Indraratna et al. (2011) have all noted that the maintenance cost of ballasted track is dominated by this cycle, and that delaying the onset of measurable degradation by even a few years has first-order economic consequences for a national rail network.

Under ordinary cyclic traffic the process of ballast degradation is gradual and reasonably predictable. Under *impact* loading—the class of loading generated at rail joints, wheel flats, transition zones where the track stiffness changes abruptly, turnouts, and level crossings—the process is accelerated by impulse stresses that rise briefly to values well in excess of the contact and tensile strength of individual grains (Remennikov and Kaewunruen, 2008). Li and Davis (2005) reported accelerated settlement rates in the metre-scale vicinity of bridge abutments that were several times the rate in the adjoining line, and attributed the acceleration directly to the sharp change in support stiffness that produced momentary contact-stress amplification. It is precisely at these hot-spots of impact loading that ballast degradation is most costly, both in maintenance and in the small-probability tail of safety-relevant events.

For Indian Railways—the fourth-largest rail network in the world by route-kilometre and one of the most heavily loaded by axle-load-times-traffic—the engineering problem of ballast degradation is particularly sharp. The network operates a mix of 22.5 t axle-load freight, heavy-haul corridors with axle loads proposed to move beyond 25 t, high-density suburban passenger services, and, increasingly, semi-high-speed passenger traffic. The ballast specification governing this mix—Research Designs and Standards Organisation (2023)—prescribes angular, machine-crushed stone with a 20–65 mm size envelope. The in-service vertical contact stress at the ballast–sleeper interface is estimated to range from approximately 240 kPa under ordinary passenger loading to values approaching 500 kPa under the combined static and dynamic action of heavy freight and impact events. Any laboratory test that purports to characterise the degradation behaviour of Indian ballast must reproduce particles in this size range and stresses in this band; any test that does not, reports a number whose physical meaning is uncertain.

1.2 Problem statement

This thesis addresses two coupled gaps in Indian railway ballast assessment—gaps that reinforce each other and cannot be closed independently.

Railroad ballast, typically composed of coarse, angular aggregates, plays a critical role in providing structural support, stability, and drainage for railway tracks (Selig and Waters, 1994). Under repeated train loading, ballast undergoes degradation through mechanisms such as particle breakage, abrasion, and fouling, leading to reduced track performance, increased settlement, and higher maintenance costs (Indraratna et al., 2019). Traditional methods for assessing ballast degradation, such as sieve analysis and manual inspection, are time-consuming, labour-intensive, and often subjective, limiting their scalability and precision. The degradation process is influenced by factors including impact-loading conditions, particle morphology (size, shape, and surface texture), and environmental



Figure 1.1. Indian Railways ballasted track in service, showing the sleeper–ballast system.

variables, yet these interactions are not fully understood or efficiently quantified (Bian et al., 2020).

The primary challenge is the lack of an efficient, objective, and automated method to assess railroad ballast degradation under realistic loading conditions. Current drop-weight impact tests simulate train-induced stresses, but standard protocols do not adequately account for variations in loading intensity, frequency, or ballast morphological change over time (Koohmishi & Palassi, 2021). Moreover, traditional post-test analysis fails to capture the dynamic evolution of particle characteristics, such as angularity loss or fragmentation patterns, which are critical to understanding degradation mechanisms (Liu et al., 2022). This gap hinders the ability to predict ballast performance, optimise maintenance schedules, and design more durable track systems. There is a need for an innovative approach that integrates advanced testing conditions with analytical tools to provide a comprehensive, data-driven assessment of ballast degradation.

1.3 Objectives of the thesis

The thesis pursues five objectives:

1. To modify the standard drop-weight impact-test apparatus (based on European standards) to replicate Indian Railways ballast characteristics and loading conditions.
2. To evaluate the degradation behaviour of railway ballast under repeated impact loading using the modified testing set-up.

3. To investigate the effectiveness of different reinforcement conditions in reducing ballast degradation under impact loads.
4. To quantify morphological changes in degraded ballast particles using three-dimensional image-processing techniques (e.g. shape, angularity, and size variations).
5. To establish relationships between impact-induced degradation and morphological characteristics for improved assessment of ballast performance.

1.4 Scope

The scope of this thesis focuses on the experimental and analytical investigation of railway ballast degradation under controlled impact-loading conditions, with particular emphasis on adapting existing testing methodologies to Indian Railways requirements. The study involves modifying and calibrating a drop-weight impact-test apparatus, originally based on European standards, to simulate realistic loading conditions representative of Indian Railways, including higher axle loads and varying traffic characteristics. Laboratory experiments are conducted to evaluate ballast degradation behaviour under repeated impact loading, examining parameters such as particle breakage, gradation change, and degradation indices. In addition, the research assesses the effectiveness of selected reinforcement or stabilisation techniques in reducing ballast deterioration. A key component of the study is the application of advanced three-dimensional image-processing methods to quantify changes in ballast particle morphology—including shape, angularity, and surface characteristics—before and after degradation. The thesis further aims to establish correlations between mechanical degradation and morphological evolution of ballast particles, contributing to an improved understanding of ballast performance. The study is limited to laboratory-scale investigations, to the specific ballast type used by Indian Railways, and to impact-loading conditions, without extensive consideration of long-term field performance or environmental effects.

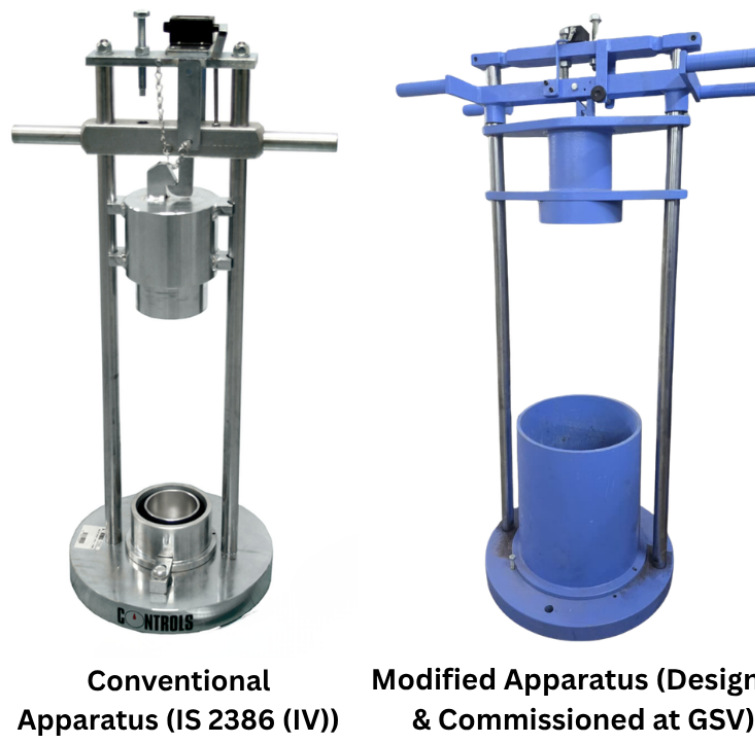


Figure 1.2. The modified impact-testing apparatus designed and commissioned at Gati Shakti Vishwavidyalaya (GSV).

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